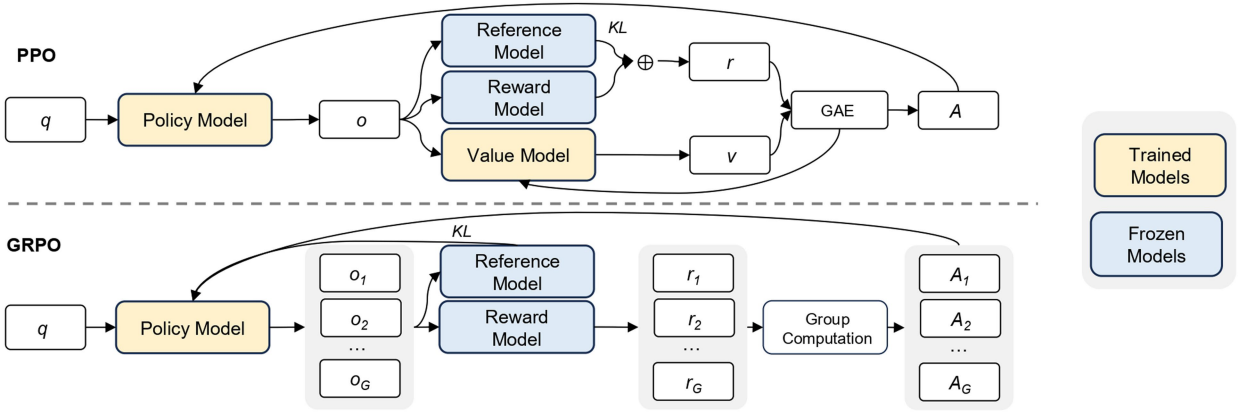

PPO
q
Policy Model
o
Reference Model
Reward Model
Value Model
KL
r
v
GAE
A
GRPO
KL
q
Policy Model
$o_1$
$o_2$
…
$o_G$
Reference Model
Reward Model
$r_1$
$r_2$
…
$r_G$
Group Computation
$A_1$
$A_2$
…
$A_G$
Trained Models
Frozen Models